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## RESULTS

# Results

01 · POISSON / NEGATIVE BINOMIAL

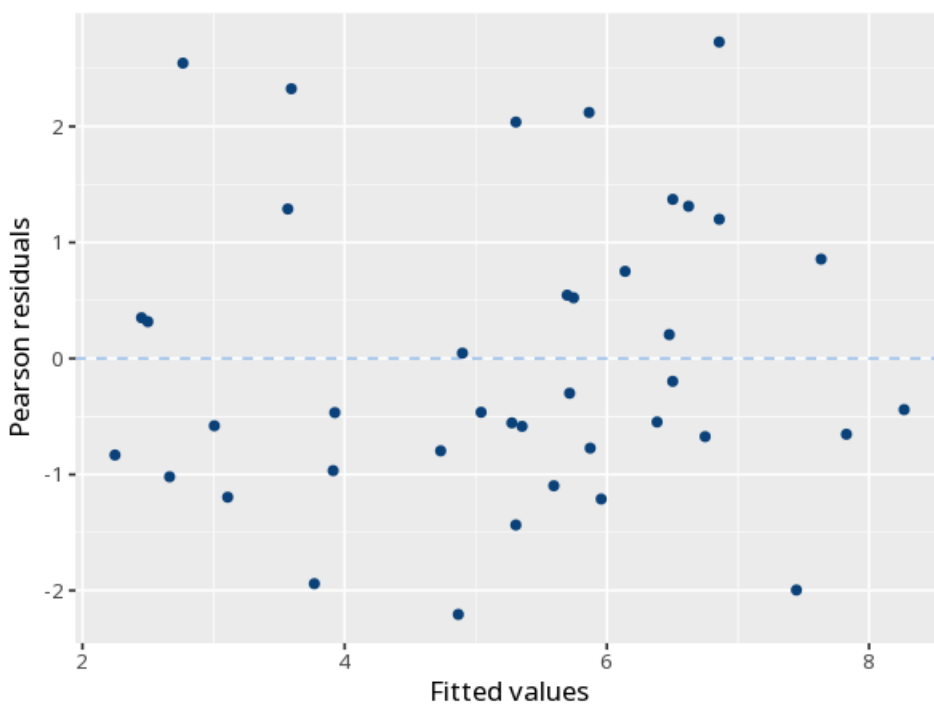
predict a count outcome

**Table 1.** Coefficients

|                   | Log-count (B)   | IRR                  |
|-------------------|-----------------|----------------------|
| (Intercept)       | −1.46<br>(0.23) | 0.23<br>[0.15, 0.36] |
| age               | 0.01<br>(0.005) | 1.01<br>[1.00, 1.02] |
| group: b          | 0.17<br>(0.14)  | 1.19<br>[0.90, 1.56] |
| Num.Obs.          | 40              |                      |
| Dispersion        | 1.69            |                      |
| Residual deviance | 67.57           |                      |
| df                | 37              |                      |
| Log.Lik.          | −98.18          |                      |
| AIC               | 202.37          |                      |
| BIC               | 207.44          |                      |

dispersion check (Poisson): if variance >> mean (over-dispersion), switch to negative binomial, which instead reports theta (its overdispersion parameter) rather than this ratio. Overdispersion test:  $\chi^2(37)$  test p = .006 — overdispersion detected.

**Figure.** Fit / residuals



## Understanding the numbers

**Incidence-rate ratio (IRR).** The multiplicative change in the expected count per one-unit increase in the predictor (>1 more, <1 fewer). Here, IRR = 1.01.

**Log-count (B).** The raw coefficient on the log-count scale that the IRR is exponentiated from. Here, B = 0.01 on the log-count scale.

**Dispersion.** Checks whether the counts are over-dispersed (variance much greater than the mean) - if so, a negative-binomial model (reporting  $\theta$  instead) fits better than Poisson. Here, dispersion = 1.69, overdispersion test  $p = .006$ .

**Residual deviance.** How much variation in the counts remains unexplained by the model - compare it to its degrees of freedom to gauge fit. Here, residual deviance = 67.57.

**df.** The degrees of freedom for the residual deviance, i.e. observations minus estimated parameters. Here,  $df = 37$ .

**Log.Lik..** The log-likelihood: how probable the observed counts are under the fitted model; it underlies AIC and BIC. Here, Log.Lik. = -98.18.

**AIC.** A fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here, AIC = 202.37.

**BIC.** Like AIC but penalizes complexity more heavily as sample size grows (lower is better). Here, BIC = 207.44.

**N.** The number of observations the model was fit on. Here,  $N = 40$ .

## How to read this test

Models event counts. Each predictor's incidence-rate ratio (IRR) gives the multiplicative change in the expected count per unit (>1 more, <1 fewer);  $p$  tests significance. Check dispersion to pick Poisson vs. negative binomial. If cases have

unequal exposure (different observation time/area/population), add an offset of  $\log(\text{exposure})$  so the model predicts rates, not raw counts — omitting it biases the IRRs. Your significance threshold ( $\alpha$ ) is 0.05.

**APA template:** Predictor age was associated with the count, IRR=1.01, 95% CI [1.00, 1.02],  $p = .007$ .

**Statistical basis:** Poisson regression (incidence-rate ratios) (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Negative binomial regression (glm.nb) for overdispersed counts (Venables WN, Ripley BD (2002). Modern Applied Statistics with S, Fourth edition. Springer, New York.); Overdispersion check (Lüdecke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D (2021). "performance: An R Package for Assessment, Comparison and Testing of Statistical Models." Journal of Open Source Software, 6(60), 3139.). Full references in the exported CITATIONS.txt.

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